Road Speed and Injury Severity in Wales

A statistical analysis of STATS19 police collision data, 2025

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Code Data

Scan for code and data

KEY FINDINGS

46% of collisions on 60 mph roads killed or seriously injured someone, against 35% across all roads

+6.3 points more likely to kill or seriously injure on a fast road than a slower road in the same council

57% of injured motorcyclists were killed or seriously injured, the highest of any road user

Introduction

Speed is a leading factor in road injury severity: crash numbers rise with speed, and fatal crashes rise fastest on rural roads (Elvik, 2013).

On 17 September 2023 Wales made 20 mph the default limit on most residential roads (Welsh Government, 2023), focusing debate on towns. We asked where serious harm now concentrates.

Research question

In Wales in 2025, are collisions on faster roads more likely to cause death or serious injury, and who is most affected?

Objectives

1. Describe when, where and who
2. Test the link between speed and severity
3. Identify who is most affected

Data and methods

- **Data:** STATS19 (Department for Transport, 2026): 3,004 collisions and 4,078 casualties, 22 Welsh councils, 2025
- **Fast roads:** 60 or 70 mph; slower roads: 20 to 50 mph
- **KSI:** killed or seriously injured; adjusted KSI corrects for police recording differences
- **Tests:** chosen after assumption checks; α = 0.05

Exploratory analysis: when, where and who

WHEN: TIME OF DAY

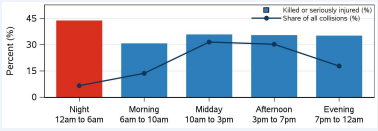


Figure 1. Bars: killed or seriously injured; line: share of collisions. Night had the fewest crashes but the most severe.

WHEN: DAY OF WEEK

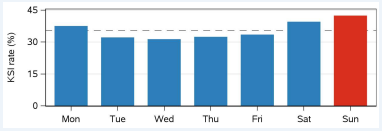


Figure 2. Share killed or seriously injured, by day (dashed line: average). Weekends were the most severe.

WHEN: MONTH

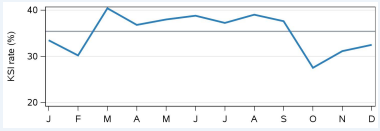


Figure 3. Share killed or seriously injured, by month (grey line: average). March to September were worse than winter.

WHERE AND WHO: KEY GROUPS

Group	n	%	KSI %
Where (collisions, n = 3,004)			
Rural	1,891	62.9	**39.8**
Urban	1,110	37.0	**27.7**
Single carriageway	2,434	81.0	**37.0**
Dual carriageway	353	11.8	**29.7**
Who (casualties, n = 4,078)			
Male	2,507	61.5	**32.6**
Female	1,567	38.4	**24.9**
Aged 17 to 24	845	20.7	**30.2**
Aged 65+	591	14.5	**35.0**

Table 1. Highest KSI rates in red.

WHERE: SPEED LIMIT

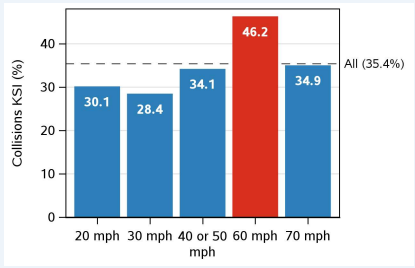


Figure 4. Share of collisions that killed or seriously injured, by speed limit (dashed line: average).

WHO: ROAD USER

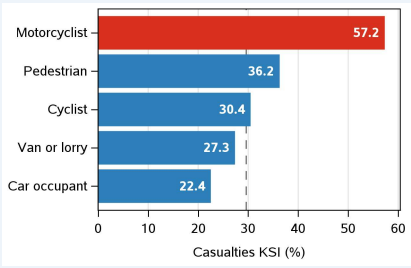


Figure 5. Share of casualties killed or seriously injured, by road user (dashed line: average, 29.6%).

Hypothesis tests: faster roads mean more serious harm

Test 1. Pearson correlation, 22 councils

H0: no correlation between a council's share of collisions on 60 or 70 mph roads and its KSI rate.

H1: a correlation exists.

Check: both variables normal (Shapiro-Wilk p > 0.5); relationship linear.

Result: r(20) = 0.59, p = 0.004 (adjusted KSI). Reject H0.

What this means: Councils with more crashes on fast roads see a larger share of people killed or seriously injured.

Test 2. Paired t test, 22 councils

H0: within councils, the mean difference in KSI rate between fast and slower roads is zero.

H1: the mean difference is not zero.

Check: differences normal (Shapiro-Wilk p = 0.11).

Result: +6.3 points (95% CI 2.7 to 10.0), t(21) = 3.59, p = 0.002. Reject H0.

What this means: Even in the same council, a crash on a 60 or 70 mph road is about 6 points more likely to kill or seriously injure.

Test 3. Mann-Whitney U, 4,028 casualties

H0: KSI and slight casualties have the same age distribution.

H1: their age distributions differ.

Check: age not normal (Shapiro-Wilk p < 0.001).

Result: median 38 vs 35 years; shift 3 years (95% CI 1 to 4), Z = 3.97, p < 0.001. Reject H0.

What this means: People who are seriously hurt tend to be slightly older, but the difference is small.

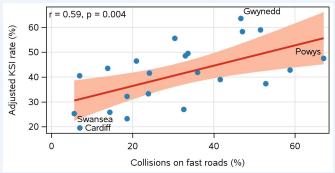


Figure 6. Each dot is one council (shaded: 95% band).

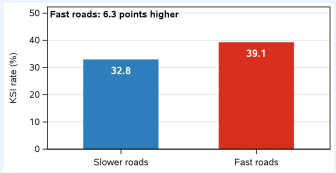


Figure 7. KSI rate: 14 of 22 councils higher on fast roads.

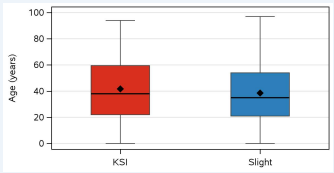


Figure 8. Age by severity (box: middle half; diamond: mean).

Discussion

Results match the Power Model: speed affects fatal crashes most on rural roads (Elvik, 2013). In Great Britain, rural roads carry 45% of traffic but 60% of deaths (Department for Transport, 2025).

Limitations: one year of police data; fast and rural roads overlap (r = 0.94); casualties in one crash are not independent.

Policy Implications

- **Beyond 20 mph:** 20 and 30 mph casualties fell 23.8% (Senedd Research, 2026); serious harm now lies on rural 60 mph roads
- **Prevention:** speed management and safer design on these roads
- **Targeting:** motorcyclist safety; council budgets where they save most lives

Conclusion

Yes. In Wales, the faster the road, the more likely a crash is to end in death or serious injury, and rural 60 mph single carriageways carry the greatest risk.

Even within the same council, a crash on a fast road was about 6 percentage points more likely to kill or seriously injure someone.

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